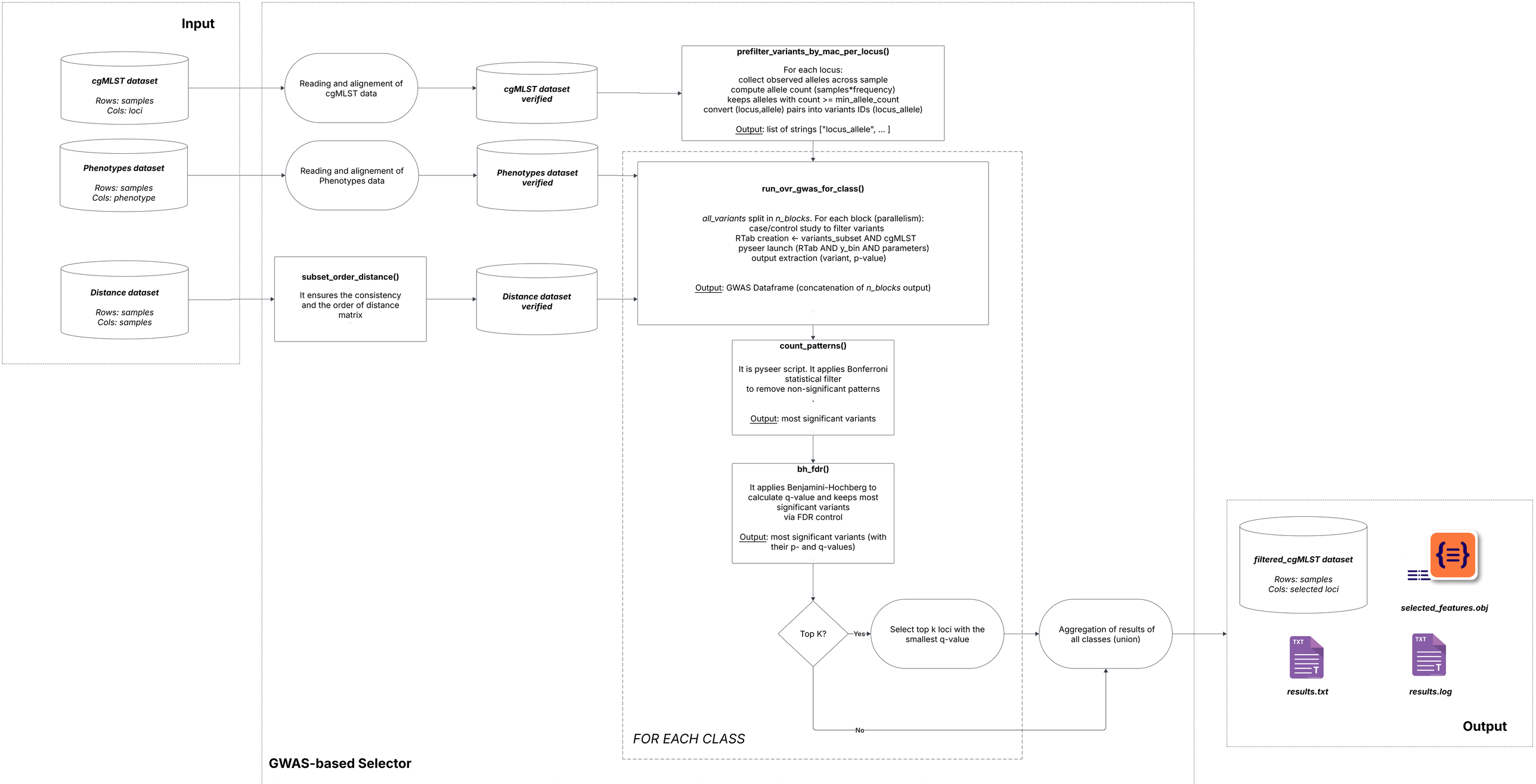
Input
cgMLST dataset
Rows: samples
Cols: loci
Phenotypes dataset
Rows: samples
Cols: phenotype
Distance dataset
Rows: samples
Cols: samples
Reading and alignement of cgMLST data
Reading and alignement of Phenotypes data
subset_order_distance()
It ensures the consistency and the order of distance matrix
cgMLST dataset verified
Phenotypes dataset verified
Distance dataset verified
prefilter_variants_by_mac_per_locus()
For each locus:
collect observed alleles across sample
compute allele count (samples*frequency)
keeps alleles with count >= min_allele_count
convert (locus,allele) pairs into variants IDs (locus_allele)
Output: list of strings ["locus_allele", ...]
run_ovr_gwas_for_class()
all_variants split in n_blocks. For each block (parallelism):
case/control study to filter variants
RTab creation ← variants_subset AND cgMLST
pyseer launch (RTab AND y_bin AND parameters)
output extraction (variant, p-value)
Output: GWAS Dataframe (concatenation of n_blocks output)
count_patterns()
It is pyseer script. It applies Bonferroni statistical filter
to remove non-significant patterns
.
Output: most significant variants
bh_fdr()
It applies Benjamini-Hochberg to calculate q-value and keeps most significant variants
via FDR control
Output: most significant variants (with their p- and q-values)
Top K?
Yes
No
Select top k loci with the smallest q-value
Aggregation of results of all classes (union)
FOR EACH CLASS
GWAS-based Selector
filtered_cgMLST dataset
Rows: samples
Cols: selected loci
selected_features.obj
results.txt
results.log
Output